

DIGITAL TO ANALOG CONVERTER

Aim:

To design and set up a 4-bit R-2R ladder DAC.

Components required:

IC's 741, 7493, resistors, CRO and trainer kit.

Theory:

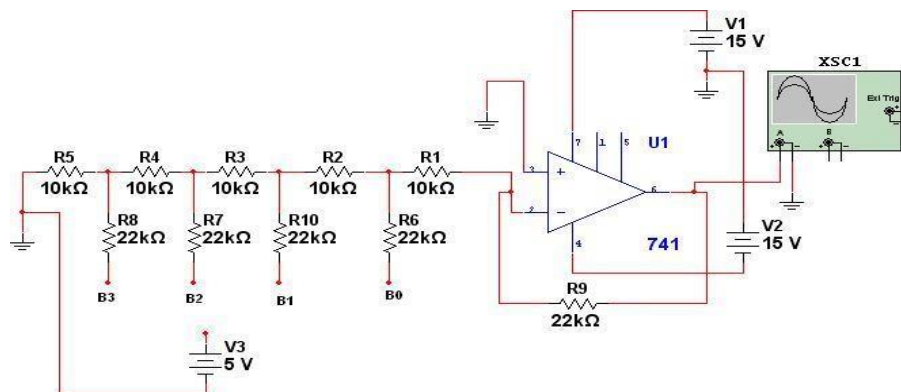
DAC is the reverse operation of ADC. The R-2R ladder DAC uses resistors of only two values R and 2R. The inputs to the resistor network are applied through digitally controlled switches from the output pins of a counter. The analog output is proportional to the value of binary input. The output is maximum when all the inputs are at logic high. The great advantage of R-2R network is that it requires only two sets of precision resistors.

The circuit diagram shown in fig1 accommodates four binary inputs. A 4-bit up counter (IC7493) outputs can be converted to provide digital inputs automatically and the analog output is observed on CRO. The magnitude of the output analogue voltage is given by the general expression

$$V_a = V_H (2^{-1}a_1 + 2^{-2}a_2 + 2^{-3}a_3 + \dots + 2^{-n}a_n)$$

Where V_H is the voltage at logic 1 state and a represent the binary states that is, 0 or 1, for example, analog output of a 4-bit R-2R DAC is

$$V_a = \frac{V_H}{2^4} (8a_1 + 4a_2 + 2a_3 + a_4)$$



Procedure:

Set up the circuit. Switch on $+V_{CC}$ and $-V_{CC}$ supplies to 741. Manually feed binary inputs 0000 through 1111 and measure the output using a multimeter and enter it in a tabular column.

Binary inputs	Analog voltage measured
0000	
0001	
0010	
.	
.	
.	
.	
.	
.	
1111	

Result:

A digital to analog converter circuit is constructed and analogue voltages for different binary inputs are measured.

